

Analyse the Impact of Zero-shot, Few-shot, and Chain-of-Thought Prompting on the Reasoning Capabilities of Large Language Models and Identify Patterns in Hallucinations and Logical Errors

Aim

To study the effect of Zero-shot, Few-shot, and Chain-of-Thought (CoT) prompting on the reasoning performance of Large Language Models using selected questions from the GSM8K dataset without writing code.

Background Theory

Zero-shot Prompting

The model is asked to solve a problem directly without any examples or intermediate reasoning hints. The prompt structure is: "Answer the following question: {question}". This method tests the model's raw generalization ability.

Few-shot Prompting

The model is given 2-3 solved examples before the actual question. This provides contextual demonstrations that guide the model on format and reasoning style. Examples: "Example 1: solved Q $\rightarrow$ A; Example 2: solved Q $\rightarrow$ A; Now solve: {question}".

Chain-of-Thought (CoT) Prompting

The model is explicitly instructed to reason step-by-step before providing a final answer. The prompt template is: "Solve the following problem step by step, showing all your reasoning, then give the final answer clearly." This reduces skip-step errors and hallucinations.

Dataset

GSM8K (Grade School Math 8K) — A benchmark dataset of grade school math word problems requiring multi-step arithmetic reasoning. Five representative questions were selected covering: egg/sale arithmetic, proportion problems, percentage increase, simple subtraction, and combined addition-subtraction.

Prompts Used

Zero-shot Prompt Template

Answer the following question: {question} Give only the final numerical answer.

Few-shot Prompt Template

Here are some example math problems: Example 1: There are 5 apples. John adds 3. How many? → 8 Example 2: A store has 20 items. They sell 7. How many left? → 13 Example 3: Maria earns \$15/hr and works 6 hours. How much? → \$90 Now solve: {question}

Chain-of-Thought Prompt Template

Solve the following problem step by step, showing all your reasoning, then give the final answer clearly. {question}

Observation Table

Model Used: Claude (claude-sonnet-4-20250514) | Date: 08-05-2026

Q.No	Question	Prompt Type	Model Response (Summary)	Actual Answer	Correct / Incorrect	Error Type	Remarks
Q1	Janet's ducks lay 16 eggs per day. She eats 3 for breakfast every morning and bakes muffins for her friends every day with 4. She sells the remainder at the farmers' market daily for \$2 per fresh duck egg. How much does she make every day at the farmers' market?	Zero-shot	\$18	\$18	Correct	None	All methods correct; simple arithmetic problem easily solved in all formats
		Few-shot	\$18	\$18	Correct	None	
		Chain-of-Thought	Step 1: Total eggs per day = 16 Step 2: Eggs eaten for breakfast = 3 Step 3: Eggs used for muffins = 4 Step 4: Eggs remaining = 16 - 3 - 4 = 9 Step...	\$18	Correct	None	
Q2	A robe takes 2 bolts of blue fiber and half that much white fiber. How many bolts in total does it take?	Zero-shot	3	3	Correct	None	Simple two-step problem; all three prompting styles produce correct answer
		Few-shot	3	3	Correct	None	
		Chain-of-Thought	Step 1: Blue fiber = 2 bolts Step 2: White fiber = half of 2 = 1 bolt Step 3: Total = 2 + 1 = 3 bolts Final Answer: 3 bolts	3	Correct	None	
Q3	Josh decides to try flipping a house. He buys a house for \$80,000 and then puts in \$50,000 in repairs. This increases the value of the house by 150%. How much profit does he make?	Zero-shot	\$120,000	\$70,000	Incorrect	Logical Error – misinterprets '150% increase' as adding 150% of original ($\$80,000 \times 1.5 = \$120,000$) without subtracting costs, or computes increase on wrong base	Zero-shot fails due to percentage interpretation error; CoT and Few-shot guide the model to correctly decompose the problem
		Few-shot	\$70,000	\$70,000	Correct	None	
		Chain-of-Thought	Step 1: Purchase price = \$80,000 Step 2: Repair cost =	\$70,000	Correct	None	

Q.No	Question	Prompt Type	Model Response (Summary)	Actual Answer	Correct / Incorrect	Error Type	Remarks
			\$50,000 Step 3: Total investment = \$80,000 + \$50,000 = \$130,000 Step 4: A 150% increase mean...				
Q4	There are 15 trees in the grove. Grove workers will plant trees in the grove today. After they are done, there will be 21 trees. How many trees did the grove workers plant today?	Zero-shot	6	6	Correct	None	Very straightforward subtraction; all methods succeed without issue
		Few-shot	6	6	Correct	None	
		Chain-of-Thought	Step 1: Initial trees = 15 Step 2: Final trees = 21 Step 3: Trees planted = 21 - 15 = 6 Final Answer: 6 trees	6	Correct	None	
Q5	Leah had 32 chocolates and her sister had 42. If they ate 35, how many pieces do they have left in total?	Zero-shot	39	39	Correct	None	Multi-step addition and subtraction; all methods produce correct answer. CoT explicitly shows the two-step reasoning.
		Few-shot	39	39	Correct	None	
		Chain-of-Thought	Step 1: Leah's chocolates = 32 Step 2: Sister's chocolates = 42 Step 3: Total chocolates = 32 + 42 = 74 Step 4: Chocolates eaten = 35 Step 5: Remai...	39	Correct	None	

Performance Summary Table

Prompting Method	Correct Answers	Accuracy (out of 5)	Common Error Pattern
Zero-shot	4 / 5	80%	Percentage/multi-step misinterpretation
Few-shot	5 / 5	100%	Generally reliable; few errors
Chain-of-Thought	5 / 5	100%	Minimal – structured reasoning reduces errors

Observations and Analysis

- Zero-shot prompting performed well on simple one-step and two-step arithmetic problems (Q1, Q2, Q4, Q5 – all correct). However, it failed on Q3 (percentage increase + multi-cost subtraction) due to misinterpretation of '150% increase', producing \$120,000 instead of \$70,000. This indicates that zero-shot struggles with multi-layered numerical transformations.
- Few-shot prompting achieved 100% accuracy. The provided worked examples helped calibrate the model's approach to framing sub-problems. The examples demonstrated how to break down costs and operations, which guided the model to correctly compute Q3's house-flipping profit.
- Chain-of-Thought prompting also achieved 100% accuracy. By explicitly showing step-by-step workings, the model avoided logical leaps and computation errors. CoT responses were the most verbose and transparent, making intermediate steps auditable.
- The most common error observed was a Logical/Conceptual Error on Q3 under Zero-shot: the model treated '150% increase' as simply 1.5x the cost rather than computing $\text{new_value} = \text{original} \times 2.5$ and then subtracting total investment.
- No hallucinations (fabricated facts or entities) were observed. Errors were purely arithmetic or logical in nature, suggesting that for math word problems, the primary failure mode is reasoning shortcut — not confabulation.
- CoT responses were more interpretable and could be manually verified, which is valuable in educational and high-stakes settings even when the final answer matches simpler methods.

Result

The experiment was successfully completed. Claude (Sonnet 4) was tested on 5 GSM8K problems across 3 prompting strategies. Zero-shot scored 4/5 (80%), while both Few-shot and Chain-of-Thought scored 5/5 (100%). The single failure in Zero-shot involved a multi-step percentage problem.

Conclusion

This experiment demonstrates that the performance of Large Language Models depends strongly on the prompting technique used. Zero-shot prompting is effective for simple, direct computations but can fail on problems requiring multiple layered calculations. Few-shot prompting improves accuracy by providing worked examples that contextualize the reasoning format. Chain-of-Thought prompting is the most robust strategy: by enforcing explicit intermediate steps, it systematically reduces logical errors and makes reasoning transparent and verifiable. Errors

observed were logical in nature (misinterpretation of percentage increase) rather than hallucinations, suggesting that structured prompting primarily mitigates reasoning shortcuts rather than factual confabulation. Therefore, for tasks requiring multi-step arithmetic or logical deduction, Chain-of-Thought or Few-shot prompting should be preferred over Zero-shot.